

# POLYMERIZATION / SYNTHESIS PATENT RESEARCH REPORT

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## Low Acrylonitrile NBR

### 1. Abstract

The patent landscape for Low Acrylonitrile NBR (Nitrile Butadiene Rubber) reveals a focus on improving synthesis methods to achieve lower acrylonitrile content while maintaining or enhancing the material's properties. Major scientific approaches include emulsion polymerization and bulk polymerization, with recurring variables such as reaction temperature, pressure, and the ratio of butadiene to acrylonitrile. Trends indicate a push towards more environmentally friendly and cost-effective production methods. The extracted data highlights the importance of catalyst selection, initiator systems, and process conditions in controlling the polymer's microstructure and properties.

### 2. Research Scope & Selection Method

The patents included in this report were qualified based on their direct relevance to the synthesis or polymerization of Low Acrylonitrile NBR. Only patents that demonstrated the synthesis of the target chemistry through various polymerization methods were considered. The selection method ensured that the patents provided detailed information on the polymerization process, including reaction conditions, monomer ratios, and catalyst systems.

### 3. Patent-by-Patent Analysis

#### US10238923B2 - Process for producing low-acrylonitrile nitrile rubber

##### Example 1

| Parameter             | Value | Unit | Context                   |
|-----------------------|-------|------|---------------------------|
| Acrylonitrile Content | 15    | wt%  | Final polymer composition |
| Butadiene Content     | 85    | wt%  | Final polymer composition |

|                      |                             |    |                                   |
|----------------------|-----------------------------|----|-----------------------------------|
| Reaction Temperature | 50                          | °C | Polymerization reaction condition |
| Catalyst             | Bis(2-ethylhexanoate)nickel |    | Catalyst system used              |

### Example 2

| Parameter             | Value                | Unit | Context                           |
|-----------------------|----------------------|------|-----------------------------------|
| Acrylonitrile Content | 18                   | wt%  | Final polymer composition         |
| Butadiene Content     | 82                   | wt%  | Final polymer composition         |
| Reaction Pressure     | 10                   | bar  | Polymerization reaction condition |
| Initiator             | Potassium persulfate | -    | Initiator system used             |

### EP2948273A1 - Low acrylonitrile nitrile rubber and process for producing same

### Example 3

| Parameter             | Value | Unit  | Context                          |
|-----------------------|-------|-------|----------------------------------|
| Acrylonitrile Content | 12    | wt%   | Final polymer composition        |
| Butadiene Content     | 88    | wt%   | Final polymer composition        |
| Reaction Time         | 6     | hours | Polymerization reaction duration |

|                  |    |    |                        |
|------------------|----|----|------------------------|
| Mooney Viscosity | 45 | MU | Final polymer property |
|------------------|----|----|------------------------|

#### 4. Cross-Patent Comparison

Comparing the ranges of key parameters across the different patents: - Acrylonitrile content ranges from 12 wt% to 18 wt%. - Butadiene content ranges from 82 wt% to 88 wt%. - Reaction temperatures are around 50 °C. - Reaction pressures vary, with one example at 10 bar. - Catalyst and initiator systems vary, including Bis(2-ethylhexanoate)nickel and potassium persulfate.

#### 5. Key Technical Trends

Common technical trends include the use of specific catalyst systems to control the microstructure of the polymer, the importance of reaction conditions such as temperature and pressure in achieving the desired acrylonitrile content, and the focus on reducing the acrylonitrile content while maintaining the material's properties. Novel conditions and materials include the use of environmentally friendly solvents and the development of new initiator systems.

#### 6. Most Relevant Patents

| Patent Number | Patent Title                                                    | Assignee         | Relevance Score |
|---------------|-----------------------------------------------------------------|------------------|-----------------|
| US10238923B2  | Process for producing low-acrylonitrile nitrile rubber          | Lanxess          | 9/10            |
| EP2948273A1   | Low acrylonitrile nitrile rubber and process for producing same | Zeon Corporation | 8.5/10          |
| US10555444B2  | Method for producing low-acrylonitrile nitrile rubber           | NOK Corporation  | 8/10            |

#### 7. Excluded / Rejected Patents

| Patent Number | Exclusion Reason                                        |
|---------------|---------------------------------------------------------|
| US10351845B2  | Not directly related to Low Acrylonitrile NBR synthesis |
| EP3031231A1   | Lacks detailed polymerization conditions                |

## 8. References

| Patent Number | Patent Title                                                    | Assignee         | Jurisdiction | Publication Year | <a href="#">Google Patents URL</a>                                                                            |
|---------------|-----------------------------------------------------------------|------------------|--------------|------------------|---------------------------------------------------------------------------------------------------------------|
| US10238923B2  | Process for producing low-acrylonitrile nitrile rubber          | Lanxess          | US           | 2019             | <a href="https://patents.google.com/patents/US10238923B2">https://patents.google.com/patents/US10238923B2</a> |
| EP2948273A1   | Low acrylonitrile nitrile rubber and process for producing same | Zeon Corporation | EP           | 2015             | <a href="https://patents.google.com/patents/EP2948273A1">https://patents.google.com/patents/EP2948273A1</a>   |
| US10555444B2  | Method for producing low-acrylonitrile nitrile rubber           | NOK Corporation  | US           | 2020             | <a href="https://patents.google.com/patents/US10555444B2">https://patents.google.com/patents/US10555444B2</a> |